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Documentation

Writing your PhD thesis

A guide to the LaTeX document class **uiophdthesis**

Dag Langmyhr (dag@ifi.uio.no)

Department of Informatics
Faculty of Mathematics and Natural Sciences

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Dag Langmyhr (dag@ifi.uio.no)

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Abstract

LaTeX is an excellent tool for writing your PhD thesis, especially in combination with the bibliography tool BibLaTeX.

The University of Oslo has defined guidelines[1] for the PhD theses, and the University of Oslo Library and the Department of Informatics have developed the document class **uiophdthesis** in accordance with these guidelines. This documentation has been typeset using that class.

[1] See https://www.uio.no/english/services/print/doctoral-thesis/index.html.

Contents

Introduction **1**
1 About the document class 3
1.1 Document structure. 3
1.2 Included research papers 3
1.2.1 Typesetting a research paper 4
1.2.2 Research paper published elsewhere 4

Paper I Practical information **5**
I.1 Installation 7
I.1.1 On a Linux computer at lfi 7
I.1.2 On your personal computer 7
I.1.3 Using Overleaf. 7
I.2 Using the document class 9
I.2.1 Document class options 9
I.2.1.1 Fonts 9
I.2.1.2 Option thumbmarks 9
I.2.1.3 Other document class options 9

Paper II An example **11**
II.1 Two other examples 17

Introduction

What is the **uiophdthesis** document class and how do I use it?

Chapter 1

About the document class

The uiophdthesis class has the following features:

- The paper size is A4, which is what the printers require (see the guidelines in footnote on page i). The thesis will be photographically reduced to 81 % when printed.
- The font size is 12 pt which will look all right in the reduced print.
- You should not attempt to create any kind of cover page. This will always be supplied by the printers.

1.1 Document structure

The standard LaTeX document structuring commands

\chapter, \section, \subsection and \subsubsection

can be used in your thesis.

You should avoid using the \part command. If you want a separation page, you should rather use the uioseparatorpage environment:

\begin{uioseparatorpage}{*title*}
some text (if you need it)
\end{uioseparatorpage}

For an example, see page 1.

1.2 Included research papers

Your thesis is likely to contain one or more research paper published in various journals. When you want to include such a research paper, you should use the command **\uiopaper**:

\uiopaper[*options*]{*title of paper*}

The available options are:

subtitle={Any suitable subtitle} provides a subtitle (if required).

authors={My Name\and Another Name\\ affiliation} is used when you want to name the author or authors.

abstract={This article describes . . . } provides an abstract for the article (if you want one on the paper title page).

compact gives a more compact title page — this is particularly useful if you have a long abstract. Page 5 has a default header page, while page 11 uses the compact version.

published={Published in *Journal of ABC*\\ additional info} for any information on the publication of the article.

info={. . . } for any additional information.

nochapter see Section 1.2.1.

1.2.1 Typesetting a research paper

If you have the LaTeX source for the research paper, you can include it here, after the \uiopaper command.

Research papers in *article* style

Many research papers were originally written in the article style, and this style has no \chapter command. To make it easier to include these in the thesis, the \uiopaper command may be given as

\uiopaper[nochapter, . . .]{*paper title*}

Then, the paper will be properly typeset with no \chapter commands but with \section as the highest structuring command.

1.2.2 Research paper published elsewhere

If you want to include the research paper exactly as it was typeset in some publication, you can import it from a PDF file using the command **\uioincludepdf** giving the PDF file as parameter.

Page numbers can often be a problem when including full-page PDF files: they might overlap those of your thesis. To remedy this, you may supply \uioincludepdf with options to alter the position of page number of the thesis. (Those of the included paper are, of course, part of the PDF file and cannot be modified.)

\uioincludepdf{*file name*} — move thesis page numbers into the margin
\uioincludepdf[numbers=low]{*file name*} — as above, but lower
\uioincludepdf[numbers=none]{*file name*} — remove thesis page numbers

Paper I

Practical information

My Name
Your Name
The University of Oslo

Abstract: This “paper” provides instructions on how to use this document class. It also describes the available document options.

(This page is an example of a default header page for published papers.)

Chapter I.1

Installation

I.1.1 On a Linux computer at Ifi

If you are processing your LaTeX document on a stationary Linux computer at the University of Oslo, you need not worry about installing the uiophdthesis document class; it is already there.

I.1.2 On your personal computer

To use this document class on your own computer (which may run Linux, MacOS or Windows) you must do the following:

1. Fetch https://www.mn.uio.no/ifi/tjenester/it/hjelp/latex/uiotheses.zip. (Click on this URL to download the file.)

2. Unzip the files. You may place all the files in the same folder as your LaTeX source files.[1]

And that should be all.

I.1.3 Using Overleaf

If you are using Overleaf (see https://www.overleaf.com) to write your thesis, you must do the following:

1. Select "New Project".

2. Under "Institution Templates", select "University of Oslo".

3. Select "UiO PhD thesis".

4. Select "Open as Template".

Once this has been done, you may use the document class.

[1] If you know where LaTeX packages are kept on your computer, you can save them there to make them generally available. Remember to refresh your file name database afterwards.

Chapter I.2

Using the document class

To use this document class, just start your LaTeX file with

\documentclass[*options*]{uiophdthesis}

I.2.1 Document class options

I.2.1.1 Fonts

The default document font for uiophdthesis is the standard LaTeX font *Computer Modern* in combination with *Helvetica*. You may select a different font by using one of these options to \documentclass:

font=garamond selects Garamond with *Helvetica*

font=noto selects Noto serif with Noto sans

font=times selects *Times Roman* with *Helvetica*

I.2.1.2 Option thumbmarks

This option provides highly visible thumb marks on the front page for all papers included using the command \uiopaper (see Section 1.2 on page 3). Pages 5 and 11 show such thumb marks.

I.2.1.3 Other document class options

All other options to \documentclass are passed on to the packages you use.

Paper II

An example

II

A Name
University of Somewhere

(This page is an example of a compact header page for published papers; see Section I.2.1.2 on page 9.)

Abstract: This “paper” shows a suitable base text for a LaTeX file for a PhD thesis, and it describes the various specifications used, line for line.

The uiophdthesis package comes with a base file named **uiophdthesis-base.tex** containing the basic layout of your thesis; see Figures II.1 on page 15 and II.2 on page 16. The idea is that you make a copy of that file, modify the specified texts, and then write your thesis.

Line 1: The document class should be **uiophdthesis**. You must also specify the language of your thesis. Include option **thumbmarks** if you want thumbmarks; see Section I.2.1.2 on page 9.

Line 2: UTF-8 is the most common character encoding in use today, so, unless you specify otherwise in your text editor, you are likely to get this encoding.

Line 3: The url package provides the \url command which is very useful for typesetting long internet addresses. These should be set in a sans serif ("sf") typeface (rather than `teletype`). For an example, see Section I.1.2 on page 7.

Lines 4–8: These packages should always be included:

babel handles language adaption.

csquotes supports quote marks in various languages. This package is required by biblatex; see below.

graphicx provides support for including illustrations.

textcomp adds many useful symbols.

varioref gives improved features for crossrefererencing.

biblatex loads BibLaTeX which handles bibliographies.[1] The package options given here are recommended; they use the numeric citation style favoured in natural science.

hyperref provides hyperlinks both internally and externally.

bookmark fixes some minor bugs in hyperref.

Lines 10–11: You must always state a thesis title. You may also provide a subtitle, but that is not mandatory.

Line 12: Don't forget you own name! If there are several authors, separate the names with an \and.

Line 14: \addbibresouce specifies the name/s of your BibLaTeX bibliography file/s.

Line 17: \frontmatter specifies the start of the thesis front matter, i.e., abstract, table of contents etc.

Line 18: prints a title page. (This will appear after the cover page so it is recommended.) Normally, just the title, subtitle and author's name is printed, but you may use these options to provide additional information:

[1] *Local guide to BibLaTeX* at https://www.mn.uio.no/ifi/tjenester/it/hjelp/latex/biblatex-guide.pdf is a simple introduction to creating your bibliography.

dept={*department name*} gives the name of the department.

fac={*faculty name*} gives the faculty name.

info={*additional info*} provides additional information.

supervisor={*supervisor's name*} names the supervisor.

supervisors={*supervisor's name*\and *name*\and *name*} gives the names of the supervisors, if there are more than one.

year={*year*} indicates the year; it is only required when the year is other than the current one.

Lines 27–29: contains your abstract.

Lines 30–33: contains your abstract in a different language, for example in Norwegian. **A Norwegian abstract is mandatory at the University of Oslo.**

Lines 35–37: produces your tables of content, figures and tables, accordingly.

Lines 39–41: is your preface.

Line 43: shows the start of the main part of your thesis.

Line 44–46: gives you a separation page.

Line 48–49: are the standard structuring commands.

Line 51–59: contains a research paper typeset as part of the thesis.

Line 61–62: contains a different research paper avaible as a PDF file.

Line 64–67: gives another separator page followed by some text.

Line 71: starts the back part containing appendices, bibliography and such.

Line 72: prints the bibliography created by BibLaTeX.

uiophdthesis-base.tex

```
\documentclass[UKenglish,thumbmarks]{uiophdthesis}   %% ... or USenglish
\usepackage[utf8]{inputenc}                          %% ... or latin1
\usepackage[T1]{url}\urlstyle{sf}
\usepackage{babel, csquotes, graphicx, textcomp}
\usepackage[nospace]{varioref}
\usepackage[backend=biber,style=numeric-comp]{biblatex}
\usepackage[hidelinks, hypertexnames=false]{hyperref}
\usepackage{bookmark}

\title{The title of my thesis}        %% ... or whatever
\subtitle{A subtitle}                 %% ... or whatever
\author{My Name}                      %% ... or whoever

\addbibresource{mybib.bib}            %% Your bibliography file

\begin{document}
\frontmatter{}
\maketitle[
  supervisor={A Supervisor},
  %% supervisors={A Supervisor\and Another Supervisor}, %% if more than one
  %% dept={Department of Something},  %% if you want to include it
  %% fac={Faculty of Theory},         %% if you want to include it
  %% info={Additional info},          %% if you need it
  %% year={2099}                      %% unless current year
]

\begin{abstract}
  Here come 3--6 sentences describing your thesis.
\end{abstract}
\begin{xabstract}[Sammendrag]               %% ... or Abstract or ...
  Here comes the abstract in a different language.
  Note that a Norwegian abstract is mandatory!
\end{xabstract}

\tableofcontents{}                          %%
\listoffigures{}                            %% (omit if none)
\listoftables{}                             %% (omit if none)

\begin{preface}
  Here comes your preface, including acknowledgments and thanks.
\end{preface}
```

Figure II.1: The file uiophdthesis-base.tex (part I)

```
\mainmatter{}
\begin{uioseparatorpage}{Introduction}  %% ... Innledning or Innleiing
  ... if you need a separator page
\end{uioseparatorpage}

\chapter{Background}                   %% ... or Bakgrunn
\section{Xxx's work}                   %% ... or whatever

\uiopaper[compact,                     %% (if you want a compact form)
  %% subtitle={A subtitle},            %% (if you want a subtitle)
  %% authors={My Name\and Your Name}, %% (name the authors; separate with \and)
  %% published={Journal name},         %% (publishing info)
  %% info={Additional info},           %% (any additional info)
  %% abstract={Article abstract},      %% (if you need one)
]{My first paper}                      %% Start another paper
\chapter{Planning the project}         %% ... or ??
Some text ...

\uiopaper{My second paper}             %% ... or ??
%% \uioincludepdf{mypaper2.pdf}        %% (PDF of a published paper)

\begin{uioseparatorpage}{Conclusion}  %% ... or ?? (if any)
  %% any information
\end{uioseparatorpage}

\chapter{Results}                      %% ... or ??
Some text ...

\backmatter{}
\printbibliography{}
\end{document}
```

Figure II.2: The file uiophdthesis-base.tex (part II)

Chapter II.1

Two other examples

- The file uiophdthesis-guide.tex shows the LaTeX source code for this documentation.
- The file uiophdthesis-demo.pdf presents a demonstration thesis showing both the LaTeX code and the result.